Lecture Notes: Substitution and Transposition Ciphers

Math F113X

Review **shift ciphers**

What is the key in the shift cipher below?

KEY: 11

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
in	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
out	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K

Substitution Cipher

A shift cipher is a particular type of **substitution cipher**.
Below is a substitution cipher that is **not** a shift cipher.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
in	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
out	E	F	G	L	8	A	R	Q	T	U	V	P	B	D	N	O	H	M

	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
in	S	T	U	V	W	X	Y	Z	0	1	2	3	4	5	6	7	8	9
out	I	1	4	7	J	6	5	K	9	3	2	0	C	X	S	Z	Y	W

1. Encrypt: MEET APRIL 3
   B881EOMTPO ← zero
2. Decrypt: F4IE19Z09
   BUSAT0730

Bus at 07:30

Substitution ciphers

- What key is needed to decrypt a message using this encryption scheme?

  A → E? tells us nothing about anything else!
  You need the entire grid to decrypt!

- Which substitution is, in general, harder to break, a shift cipher or one that is not a shift cipher?

  non-shift substitution!

- What strategies would you use to try to break a substitution cipher that is not a shift cipher?

  Frequency Analysis: match frequently occurring letters in the ciphertext with the most frequent letters in English

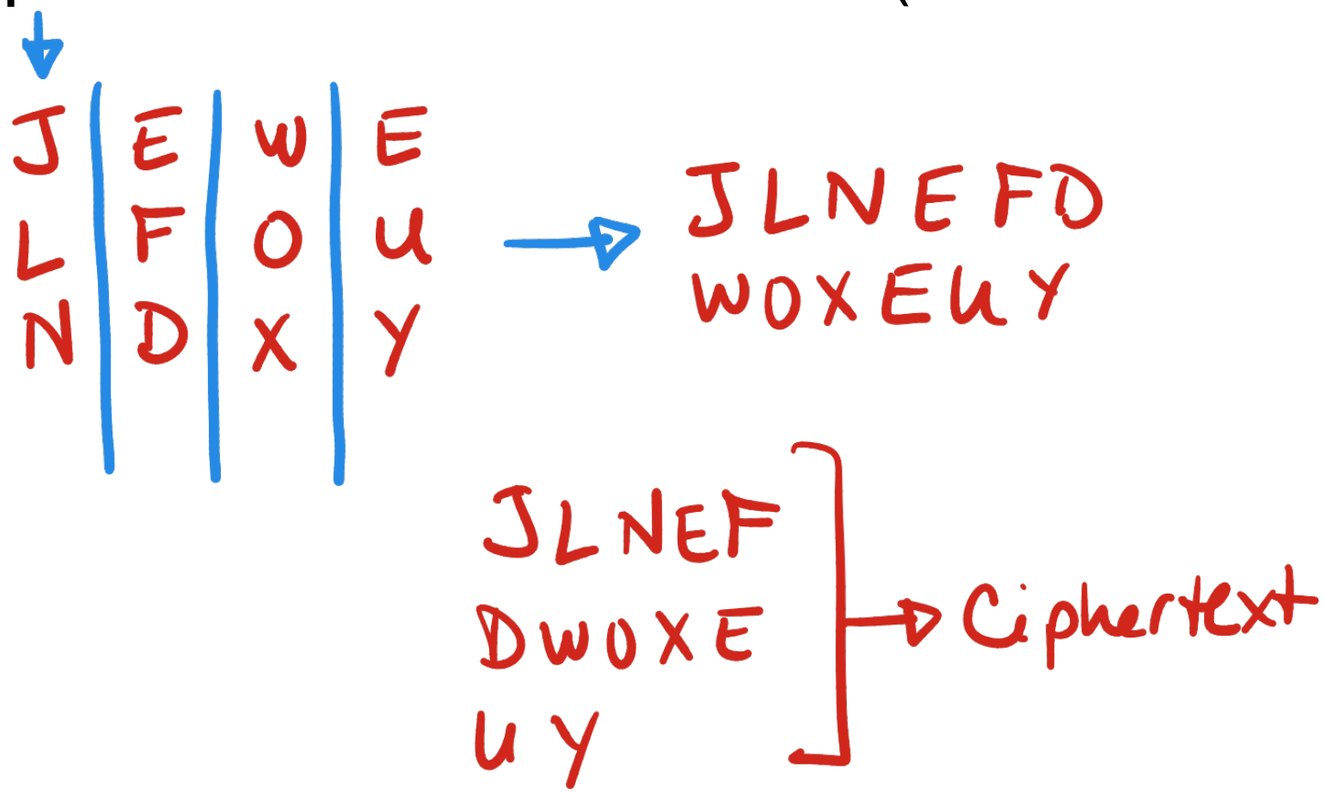
Transposition Cipher: Encrypt

Another encryption scheme is called a **transposition cipher**. Encode

JEWEL FOUND

using a transposition cipher of rows of 4 letters (4 columns).

1. Put in rows
2. read out columns



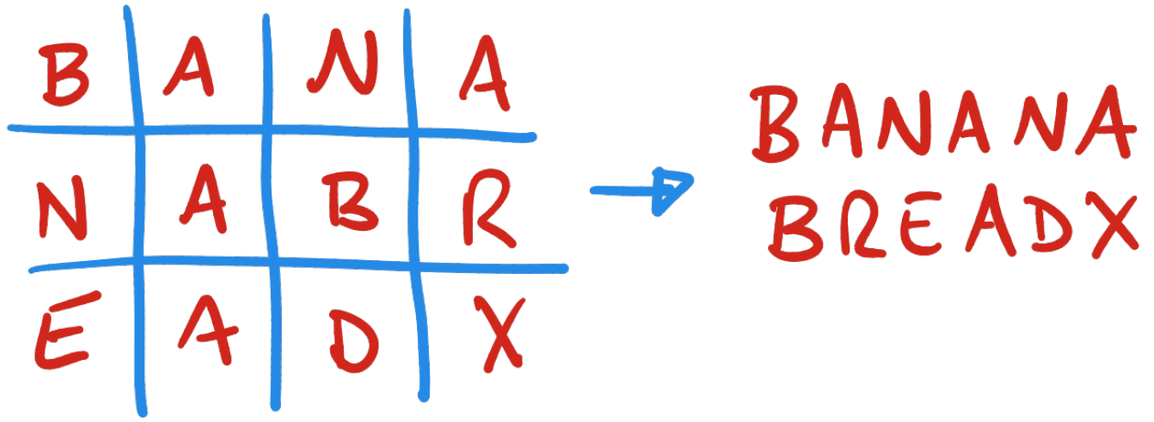
Transposition Cipher: Decrypt

Decrypt

BNEAA ANBDA RX ← 12 characters

if it was encoded with a transposition cipher with rows of 4 letters (4 columns). 12/4 = 3 ← # rows

1. How many rows?
2. Put in columns
3. read out rows



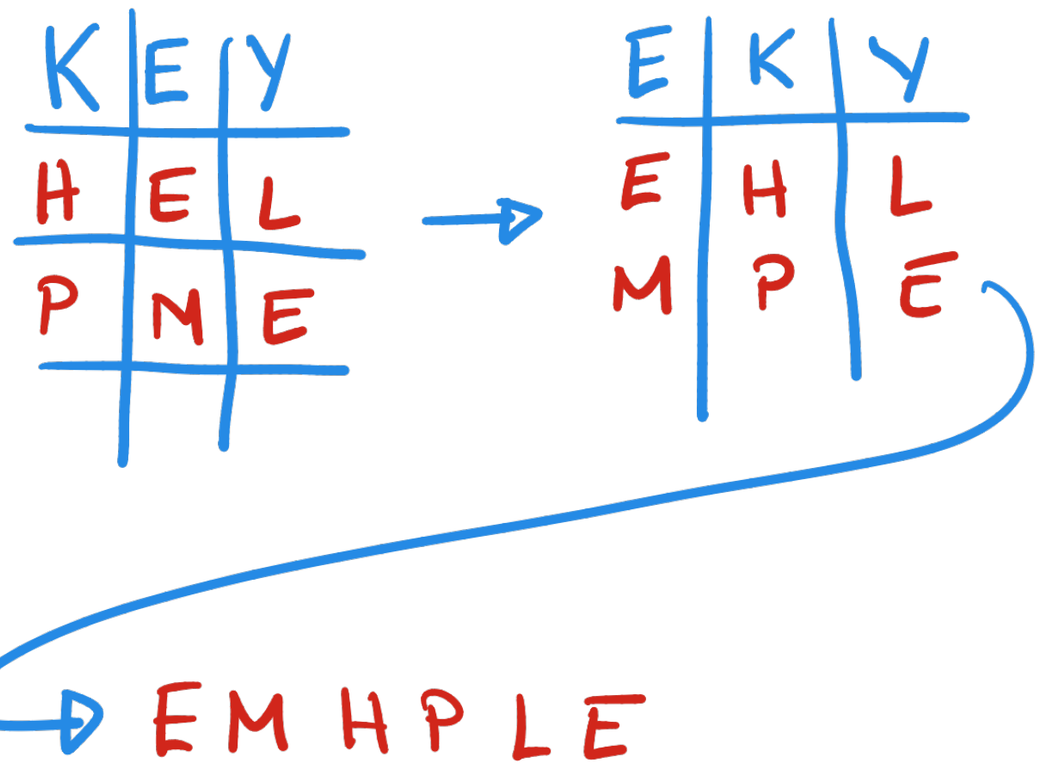
Transposition Cipher with key: encrypt

Encrypt using a transposition cipher with a keyword KEY:

H E L P M E

KEY → EKY

1. Put in grid by rows
2. Mix up columns
3. Read out columns



Don't choose a keyword with repeated letters!

Transposition Cipher with key: decrypt

Decrypt using a transposition cipher with a keyword KEY:

F N T I U I O D X ← 9 characters

1. Determine how many rows $9/3 = 3$ ← rows
2. Put in grid by **columns**, labeled by SORTED keyword
3. un-mix-up columns KEY → EKY
4. Read out **rows**

